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COMMENTARY

Clinical Decision Support: Moving Beyond Interruptive "Pop-up" Alerts

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We cannot afford to commit genericide with clinical decision support (CDS). *Genericide* refers to when a product or service becomes so widely used that consumers refer to generic versions with the brand name. Copying a document is xeroxing. Searching for information on the internet is googling. Digitally altering an image is photoshopping. Videoconferencing is zooming. Clinical decision support, however, is so much broader than the narrow offering of a "best practice advisory"—an interruptive "pop-up" triggered by an if/then criterion—and should not be genericized as such. Clinical decision support can also electronically deliver patient-specific recommendations in less obtrusive ways as reminders, information display, info buttons, order sets, documentation templates, algorithms, and calculators. Given the pervasiveness of interruptive pop-up alerts, health care workers tend to think of them as one and the same as CDS—overriding pop-ups up to 96% of the times they fire without taking action or even recognizing the recommendation being offered.[1] These interruptions are not without consequence. They necessitate task switching, which in turn increases cognitive load and introduces the opportunity for error and task abandonment, both of which threaten patient safety.[2] In situations in which pop-up alerts are dismissed, CDS is not having its intended consequence, probably because of a failure to adhere to the "5 rights" of CDS: delivery of the right information to the right person in the right format through the right channel and at the right time in the workflow.[3] Here, we present evidence of 2 interventions that have done this and the technology that will allow similar solutions to become more commonplace, thereby displacing ineffective pop-up alerts.

To date, CDS has had limited and at times unintended effects on outcomes with implementation rarely optimized for workflow unless it has been designed with rigorous end user evaluation.[4-7] One study, a 2020 meta-analysis across 122 randomized trials of CDS including more than 1.2 million patients, found that CDS systems were associated with a less than 6% increase in average number of patients achieving the desired outcome.[8] The authors speculated that CDS effectiveness was markedly limited by alert fatigue and poor workflow integration. The rapid proliferation of CDS in routine practice without rigorous evaluation of its implementation and impact on intended (and unintended) outcomes can promote genericide of interruptive pop-up alerts in place of CDS that truly supports clinical decision-making. Indeed, although commonly used to comply with regulatory or national quality metrics, pop-up alerts are typically ineffective. The failure to optimize CDS for routine clinical workflows can lead to unintended *harm*. In a 2021 patient-level randomized trial of 6030 adult inpatients with acute kidney injury at 6 hospitals, an interruptive alert was repeatedly delivered to all clinicians on the care team each time they opened the patient's chart with the option to open an order set of best practices for kidney injury management. Despite increased fluid administration in the alert group, there was no observed effect on progression of kidney injury or dialysis. More concerning was that at nonteaching hospitals, the risk of death in the alert group was higher than in usual care (relative risk, 1.82; 95% CI, 1.22 to 2.72; P=.006).[9]

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Rigorous user-centered optimization of CDS before implementation or summative evaluations of barriers to implementation and effectiveness may have illuminated potential drivers of this observed difference in mortality risk and potentially averted harm.

Interruptive alerts must trigger on activities within the electronic health record (EHR), such as opening orders or dispositioning the patient. Whereas they may contain the "right information," they rarely trigger at the right time. This is likely to be an artifact of the silo design of EHR interfaces with tab-based activities for reviewing results, prior notes, and past medical history and placing orders as discrete activities when, from the perspective of decision-making, they are interconnected. Therefore, the sociotechnical context of CDS implementation matters for timing, and success is dependent on deliberate design of CDS with user input. So, what should the next generation of CDS look like to address these barriers? As evidence of CDS evolution, our large academic health system has implemented two programs with promising results: Emergency Department–Initiated BuprenorphinE for Opioid Use Disorder (EMBED) and Care Signature pathways.[10,11]

EMBED is a clinician-facing CDS tool implemented after in-depth formative evaluation to optimize integration into clinical workflows to support diagnosis of opioid use disorder, withdrawal assessment, and appropriate buprenorphine administration and prescribing—a complex, unfamiliar practice to many emergency clinicians.[12,13] To facilitate and to streamline decision-making, EHR activities including clinical documentation, orders, prescriptions, discharge instructions, and referral are automated within a single workflow. A recently completed trial of EMBED among 599 attending emergency medicine physicians caring for 5047 adult patients with opioid use disorder found a 10% increase in at least 1 buprenorphine initiation among physicians in the intervention arm compared with the control group (44.4% vs 34.0%; adjusted odds ratio, 1.83; P=.01).[10]

The Care Signature program was established in August 2020 and uses pathways to identify sources of unnecessary variation in clinical care and to delineate best practice through evidence review, expert consensus, and process improvement. All resources necessary to deliver each condition-specific best practice standard are harnessed into clinical pathways integrated into the EHR. Whereas pathways have been discussed in a variety of settings, this program implements dynamic pathways within the EHR that display relevant data, allow 1-click orders, and link to other areas of the EHR. However, these are not solely order delivery tools but rather best practice guidelines curated with representatives from quality and safety, pharmacists, social workers, health equity experts, and patient representatives to ensure that highest quality, continually updated clinical content is delivered to the user. Implementation of an emergency department–specific Care Signature pathway to support the treatment of 6600 patients thought to have COVID-19 was associated with 2.44 (1.69 to 3.55) higher odds of first-dose corticosteroid treatment compared with emergency departments without the pathway. After implementation of the pathway at all sites, eligible patients had 10.4 (8.85 to 12.2) higher odds of first-dose corticosteroid treatment compared with nonpathway users. Additional process metrics demonstrated statistically significant improvements in minutes to corticosteroid administration (54.3 [95% CI, 63.7 to 44.9]; P<.0001) and admission time (71.9 [80.6 to 63.2]; P<.0001) compared with nonpathway users.[11]

So, what makes these CDS interventions different? In contrast to interruptive alerts that increase cognitive load by task switching, EMBED and Care Signature pathways are both noninterruptive and passive decision tools that allow clinicians to access decision support *when they want to access it*. Both Care Signatures and EMBED prepopulate order fields (eg, dosing, indications), which reduces variation, cognitive load, mouse clicks, and potential for errors such

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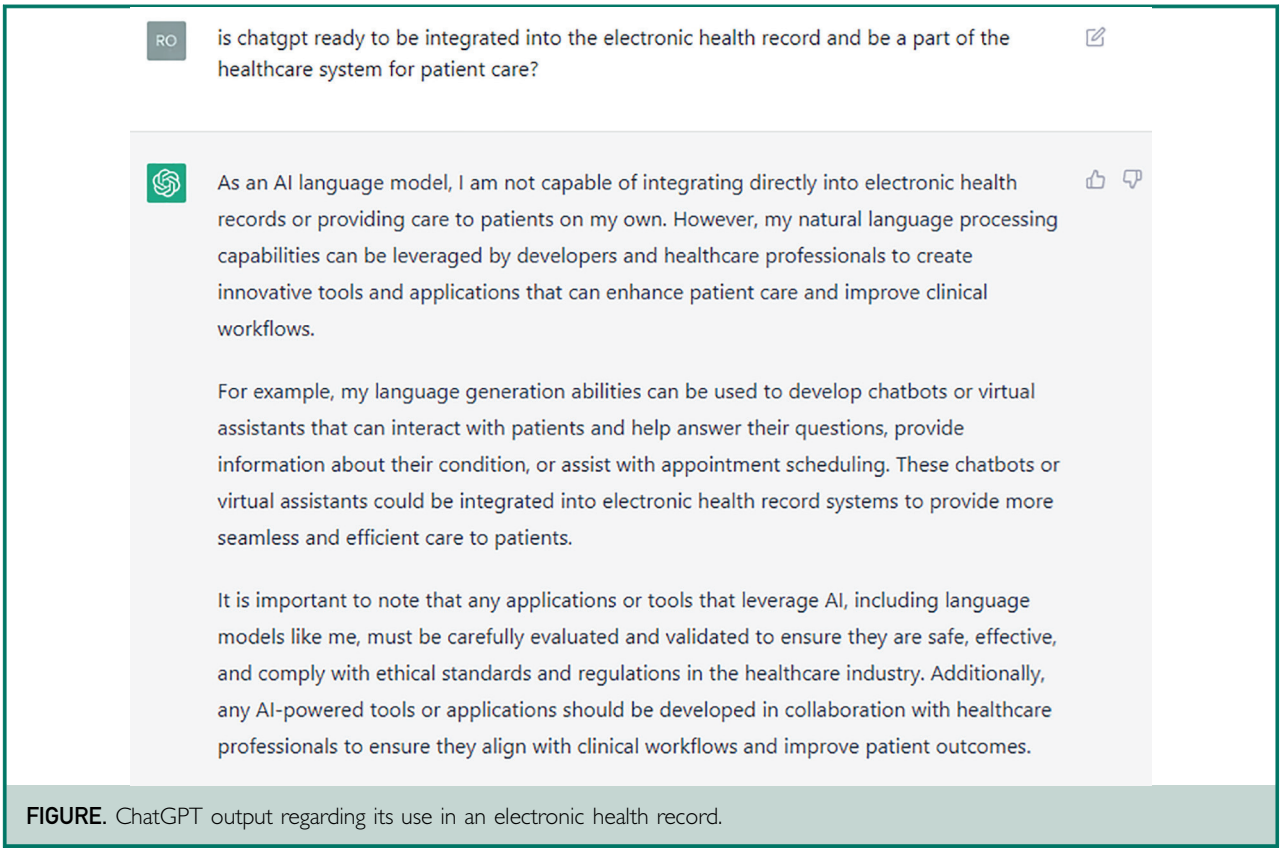


FIGURE. ChatGPT output regarding its use in an electronic health record.

as wrong imaging test or wrong dose. For example, Care Signature pathways use clinical information to display recommended pathways passively on the side of the patient EHR window but also allow clinicians to choose if and when they click to engage with the CDS. With each use of this platform, the clinician has indicated a willingness and readiness to engage with the CDS despite the multitude of competing tasks in their sociotechnical context. Ensuring the clinician's undivided attention in this way yields the best opportunity for effectiveness. These interventions require user initiation, which may not always be an option in situations involving high risk or requiring mandated compliance. However, there are emerging solutions for these situations as well, such as more carefully targeted, noninterruptive, and modal alerts.[14]

Clinical decision support users and implementers should look optimistically toward the future. More than a decade ago, the Health Information Technology for Economic and Clinical Health (HITECH) Act nurtured the adoption of EHRs, and now the 21st Century Cures Act promotes interoperability. This legislation encourages an open marketplace through platforms that can meet the specific needs of clinicians across a range of specialties. It is this interoperability that has allowed solutions like EMBED and Care Signature to emerge. With the rise of Substitutable Medical Applications and Reusable Technologies (SMART) and Fast Healthcare Interoperability Resources (FHIR), innovative platforms can now present information in ways more palatable for clinicians.[15] Fundamentally, SMART on FHIR will allow more third-party applications to interact with the EHR and to curate information from varying parts of the EHR to present a seamless user experience in streamlined ways that approach the 5 rights.

Briefly, SMART on FHIR leverages a common data language to allow 2-way communication of clinical information. Whereas external applications may have proprietary ways of managing, displaying, or interacting with patient data, using this common language allows more seamless interaction with the core EHR. Ideally, tools like this will reduce navigation time and number of clicks while highlighting relevant clinical information during the delivery of care. Currently, such tools are likely to be passive as the complexity of workflows has yet to be fully grasped. Specifically, EHRs allow multiple ways to do the same task (eg, access results, write notes, place orders) that makes it difficult to determine exactly when a clinician would be ready to see an interruptive alert. In addition, the nonlinear nature of routine care means that clinicians may interact with the EHR before or after evaluating the patient and may counsel patients before or after reviewing results, thereby creating further challenges to appropriate timing of interruptive alerts. Despite this, as passive tools, their purpose will be to provide up-to-date CDS that is available when the clinician is ready to engage, which can be at various points in a clinical encounter.

More recently, artificial intelligence (AI) and machine learning software have begun to establish themselves in the public domain. ChatGPT is one tool that has shown remarkable ability to process language—even successfully passing the medical board exams.[16] As SMART on FHIR standards become more robust and integration with the EHR more commonplace, AI tools could be deployed as surveillance measures to promote diagnosis, to improve patient quality and safety, and to promote personalized medicine.[17] This promising future may be in reach, but first, in ChatGPT's own words, "any applications or tools that leverage AI, including language models like me, must be carefully evaluated and validated to ensure they are safe, effective, and comply with ethical standards and regulations in the healthcare industry. Additionally, any AI-powered tools or applications should be developed in collaboration with health care professionals to ensure they align with clinical workflows and improve patient outcomes" (Figure).

CONCLUSION

Whereas CDS has come a long way since its inception, the interruptive nature, lack of workflow congruence, and lack of consistent positive outcome data have given CDS a bad reputation. But this does not have to persist. Our health system experience illustrates that novel platforms invoking user-centered design, such as EMBED and Care Signature, can lead to better EHR performance and patient care. ChatGPT demonstrates that natural language processing and AI tools may be deployed with success. From an administrative view, adjusting clinician behavior can be challenging, and both EMBED and Care Signature nudge clinicians to adjust behavior by articulation of best practice and automation of orders and documentation to save time and cognitive load. The rise of SMART on FHIR promises to promote interoperability, innovation, and better adherence to the 5 rights, all while promoting best practice. In this way, we can avoid genericide with CDS through a deliberate approach to design and evaluation while improving clinician workflow.

POTENTIAL COMPETING INTERESTS

The authors report no competing interests.

Abbreviations and Acronyms: **AI,** artificial intelligence; **CDS,** clinical decision support; **EHR,** electronic health record; **EMBED,** Emergency Department–Initiated BuprenorphinE for Opioid Use Disorder; **FHIR,** Fast Healthcare Interoperability Resources; **SMART,** Substitutable Medical Applications and Reusable Technologies

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